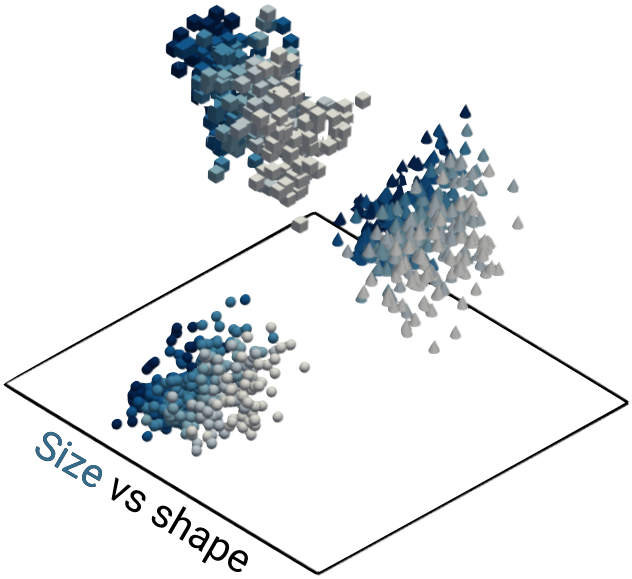

Figure 27. **Geometry of *datapoints* in OpenCLIP ViT-L/14.** We show the span of the joint features of OpenCLIP ViT-L/14.

I.4. Experiments using text encoders as probes (zero-shot results)

In the main text (Section 5.1), we analyzed the factors of the models by training linear probes on the image embeddings using gradient descent with cross-entropy. This was done for two reasons: (1) to handle concepts that are difficult to express as text prompts (e.g., visually complex backgrounds or continuous attributes like size or orientation), and (2) to avoid potential misalignment between the text and vision modalities, where the text encoder must accommodate many visual categories, potentially leading to suboptimal performance for certain domains. Here, we ask what happens when we do not take into account these problems and instead rely on the linear probes that the text encoder already produces.

In this section, we provide analogous analyses to those in the main text, but using the text encoder as probes instead of external linear probes for two datasets: PUG-Animal and ImageNet-AO. We use these datasets for two reasons: (1) their concepts and values map naturally to text prompts, and (2) the datasets were released after the CLIP models and exhibit many unnatural concept combinations unlikely to have appeared in text captions during pre-training, and not present in the visual training data.

I.4.1. Experiments on PUG-Animal

Setup. Four concepts are exposed: character, background, scale, and texture. For each character we parse the character name into a set of words and use prompts of the form "A picture of a <character>". For each background, we use prompts of the form "A picture of a <background>" (detailed in Tab. 4).

We map numeric scale values and texture labels to descriptive prompt templates for evaluating the models. Specifically, for scale, we use:

- `0.7` → "A picture of a small object"
- `1.0` → "A picture of a medium-sized object"
- `1.3` → "A picture of a large object"

For textures, we use the following mappings:

- "Sky" → "A picture of an object in sky texture"
- "Grass" → "A picture of an object in grass texture"
- "Asphalt" → "A picture of an object in asphalt texture"

These prompt templates are used to generate the corresponding text embeddings for each concept, matching exactly with the setup of the experiments in the main text.

Table 4. **Mapping from class names to clean prompt names for PUG-Animal experiments.**

Original Name	Prompt Name
Desert	a desert
Tableland	a tableland
EuropeanStreet	a European street
OceanFloor	the ocean floor
Racetrack	a racetrack
Ruins	ancient ruins
TrainStation	a train station
BusStationInterior	the interior of a bus station
BusStationExterior	the exterior of a bus station
IndoorStairs	indoor stairs
Circus	a circus
BoxingRing	a boxing ring
Mansion	a mansion
ShoppingMall	a shopping mall
ConferenceRoom	a conference room
VillageOutskirt	a village outskirt
VillageSquare	a village square
Courtyard	a courtyard
Forge	a forge
Library	a library
Museum	a museum
Gallery	an art gallery
Opera	an opera house
Restaurant	a restaurant
RuralAustralia	rural Australia
AustraliaRoad	a road in Australia
ShadyRoad	a shady road
SaltFlats	salt flats
Castle	a castle
Temple	a temple
Snow	a snowy landscape
Grass	a grassy field
DryGrass	a dry grassland
Forest	a forest

Concretely, for each concept value $j \in [n]$, we pass the prompt template through the text encoder g to obtain a (ℓ_2-normalized) probe vector $\boldsymbol{w}_{i,j} = g(p_{i,j}) \in \mathcal{Z}$, as detailed in Section 3.4.

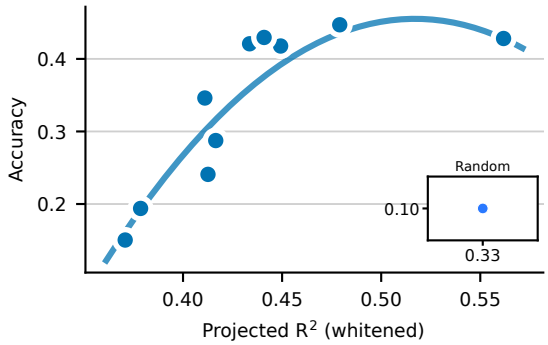

Figure 28. **Projected R^2 vs accuracy on PUG-Animal across models.** Higher projected R^2 coincides with higher accuracy on the full dataset. The probes are extracted from the text encoder.

Linearity of factors and generalization. We show the projected R^2 and average accuracy on all concept combinations on PUG-Animal across models in Fig. 28 when using the text encoder as probes. Models exhibiting higher linearity of representations generally exhibit higher accuracy on the full dataset. This coincides with the observations in the main text (Section 5.1); random baseline achieves low projected R^2 and accuracy.

Orthogonality of the factors. For each of the concepts, we compute the linear factors as detailed in the main text (Section 5.1) with the text encoder as probes. We compute the within- and across-concept orthogonality as detailed in Section I.2 and illustrate the results in Fig. 29 for each of the models.

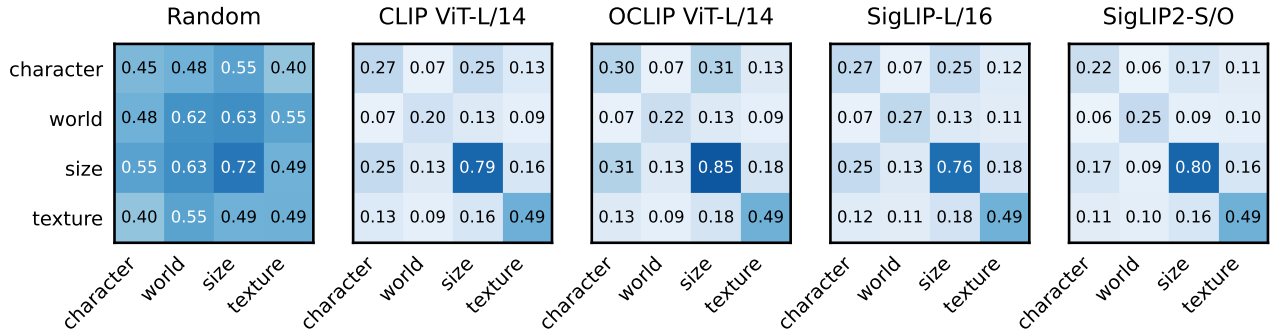

Figure 29. **Orthogonality of the factors on PUG-Animal.** Heatmaps show pairwise cosine similarity between factors for the four PUG-Animal concepts (character, world, size, texture) across multiple models. The factors are more orthogonal across concepts (off-diagonal) than within concepts (diagonal). The random baseline does not generally show this pattern.

For all evaluated models, we observe the same orthogonality pattern: the factors are more orthogonal across concepts (off-diagonal) than within concepts (diagonal). The average cosine similarity for the random baseline is higher (around 0.5) both within and across concepts.

We also note the qualitative similarity between the factors to the case when probes were trained on 90% of the concept combinations (Fig. 24, second row).

Qualitative examples. We illustrate some of the highest- and lowest-scoring samples in terms of R^2 for the SigLIP2 model in Fig. 30. We note that high-scoring samples generally depict clean scenes where the character and its size and texture are easier to discern compared to the lower-scoring samples.